

xMCF

ISO/PAS 8329

Description of mechanical connections
and joints in structural systems

ISO/PAS 8329 xMCF WG meeting
focus group on issues [#105](#) / [#61](#)
2026-06-12

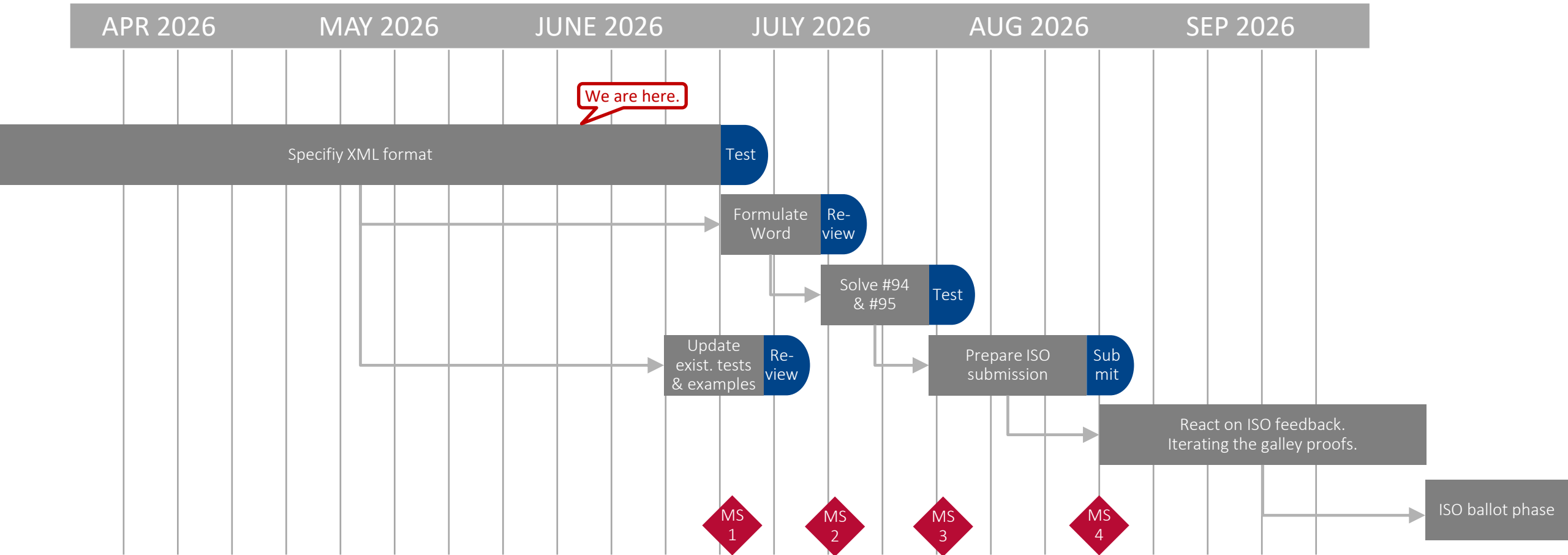
Agenda

As announced at https://github.com/economidis-nick/createXSDforxMCF/blob/master/VDA_FAT_AK_25/Meetings/2026-06-12_VideoConference/README.md

- Welcome & Flying over today's topics
- Timeline for next release
- Work in progress
- Focus on XML / DOC suggestions, XSD- / Schematron-definition & examples
- Steps we should take next



Timeline for the next release

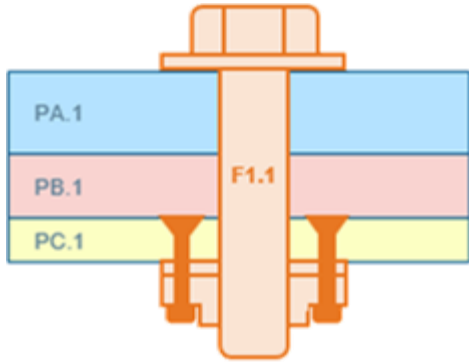


Updating the PAS (or else striving for an IS, directly) shall be cross-checked with the chair of ISO/TC 184 / SC 4.
Question was repeated at April 28th SC4/WG12 meeting. Though, no answer yet.

Issue #106: Support complex rivet-nut fasteners

Due to the composition of today's attendance list, the focus was placed on issue #106.

The task is to describe a design like this:



The discussion showed that this type of modelling is feasible. It can be described using the published standard.

GH will add a note to STEP's recommended practices and also post a comment on issue #106. The issue will then be closed.

GH outlines a solution and invites discussion:

1. The nut is riveted to part PC.1.
⇒ One `connection_group` between PC.1 & nut.
2. The bolt connects PA.1, PB.1, and PC.1.
The nut is modeled "`fixed_to`".

Conclusion of the Investigation

Combined XML verification by XML schema plus Schematron provides:

- Checks for several additional aspects, e.g. type confusion.
- Chance to separate business logic ($\mapsto$ Schematron) and formal syntax ($\mapsto$ Schema)

⇒ There are indisputable technical reasons to switch χ MCF 3.2 validation to a Schematron-based method.

However:

- Implementing the complete Schematron file will take weeks.
- Many checks in XML schema need to be reduced from “mandatory” to “optional”. Will also take time.

NE & SH volunteer for both these items.

CF will care for updating the existing χ MCF 3.2 example files. — Pending.

NE's recent comment to issue #105

- Postponed ...



Contentual Action Items $\triangleq$ Next Steps:

▪ Standard document (Word file):

- Description of overall concept & objectives, intended application & motivation etc.
- Description of semantics for internal catalog
- Description of semantics for reference to a connection template
- Definition of example use cases and in-document XML-examples

▪ XML & XSD — See discussion in [#105!](#)

- Sketching XML example file(s), incl. (w.i.p.)
 - precedence rule (w.i.p.) and dropped mandatory attribute,
 - internal (w.i.p.) and external file-based catalog
- Description of syntax for internal catalogs (w.i.p.)
- Description of syntax for reference to a connection template:
 - internal (w.i.p.), external file-based, external database-based

- Drafting XML schema (XSD) and test cases / test example files

Homework: Collect use cases which can yield e.g. test cases, shared as Markdown files in GitHub at https://github.com/economidis-nick/createXSDforxMCF/tree/issues_61_94_95_105/VDA_FAT_AK_25/Meetings/2026-02-18_VideoConference

- Investigate xMCF elements and attributes for eligibility for template.
⇒ we need a rule, here! ⇒ Homework! (cf. Tables 35—79 of the standard)

pending

pending

Next meetings

- 2026-06-12, 16:00 CEST — focus group on [#106](#)
 - objective: agree on XML suggestions & examples
- 2026-06-17, 16:30 CEST — focus group on [#105](#) / [#61](#)
 - objective: agree on XML suggestions & examples
- 2026-06-24, 16:30 CEST — focus group on [#105](#) / [#61](#)
 - objective: agree on XML suggestions & examples

CF sent / will send out invitations.

